

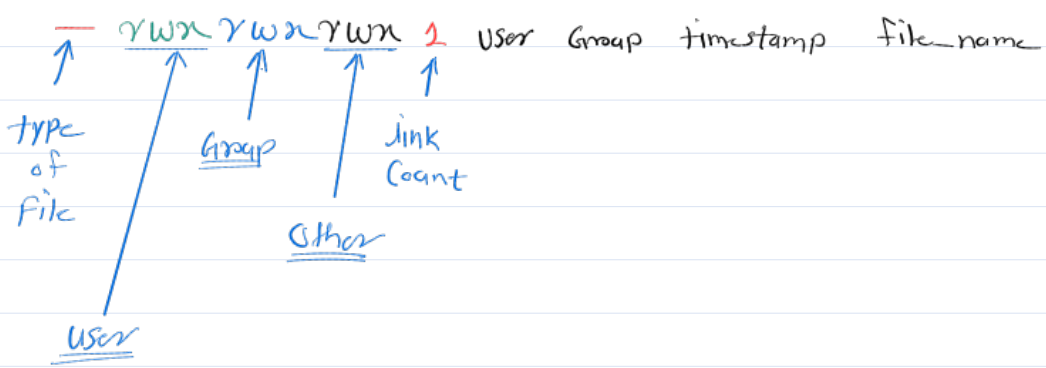
Concepts of Operating Systems And Linux Operating Systems

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✳ Types of File

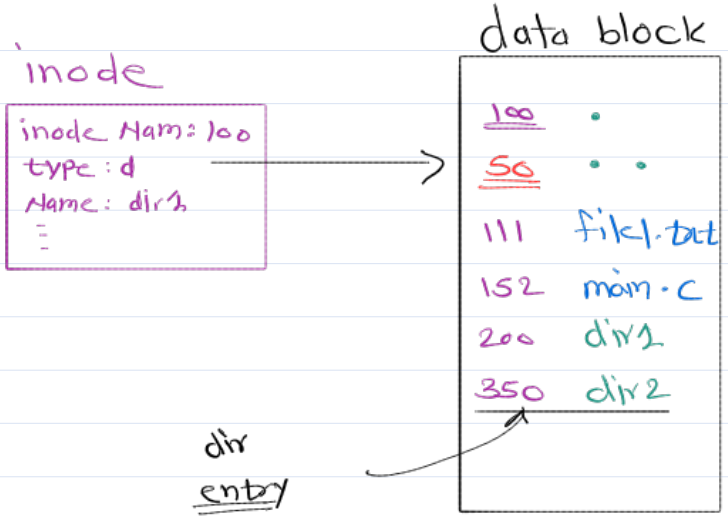
1. Regular (-)
2. Directory (d)
3. link (l)
4. Pipe (p)
5. Socket (s)
6. character (c)
7. Block (b)

ls -l

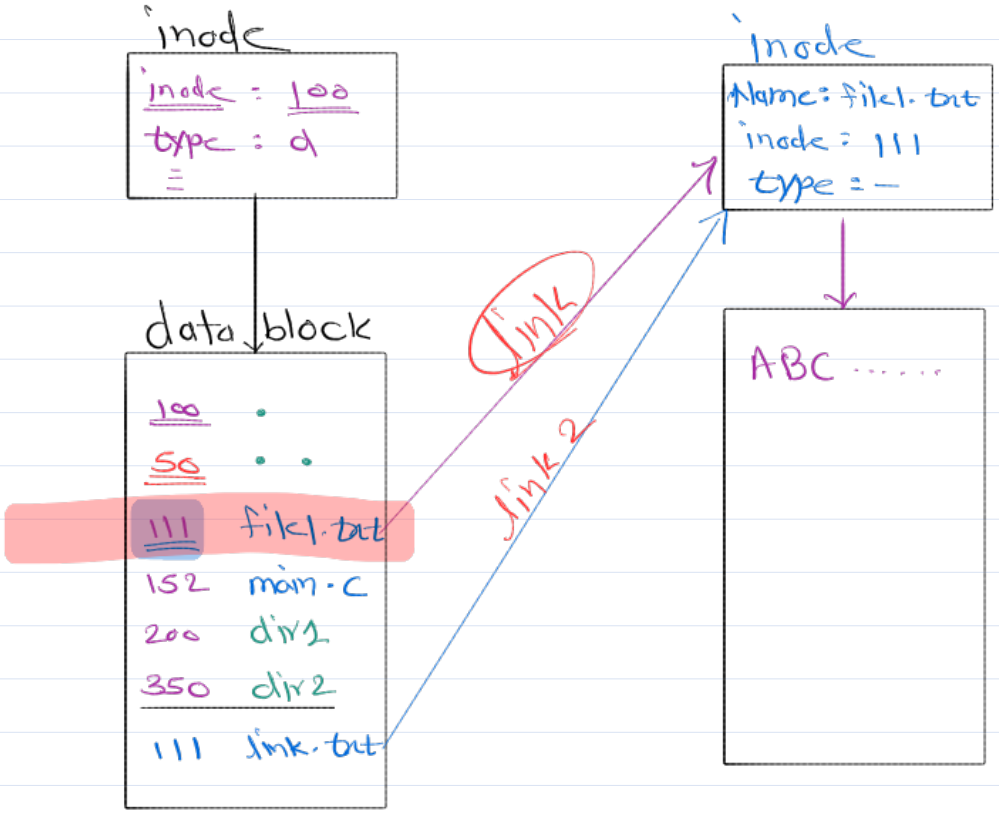


Directory file (d)

- In Linux directory is special type of file.
- Dir file also get inode and data block
  - inode store file info
  - Data block store dir-entry



Hard link - ln



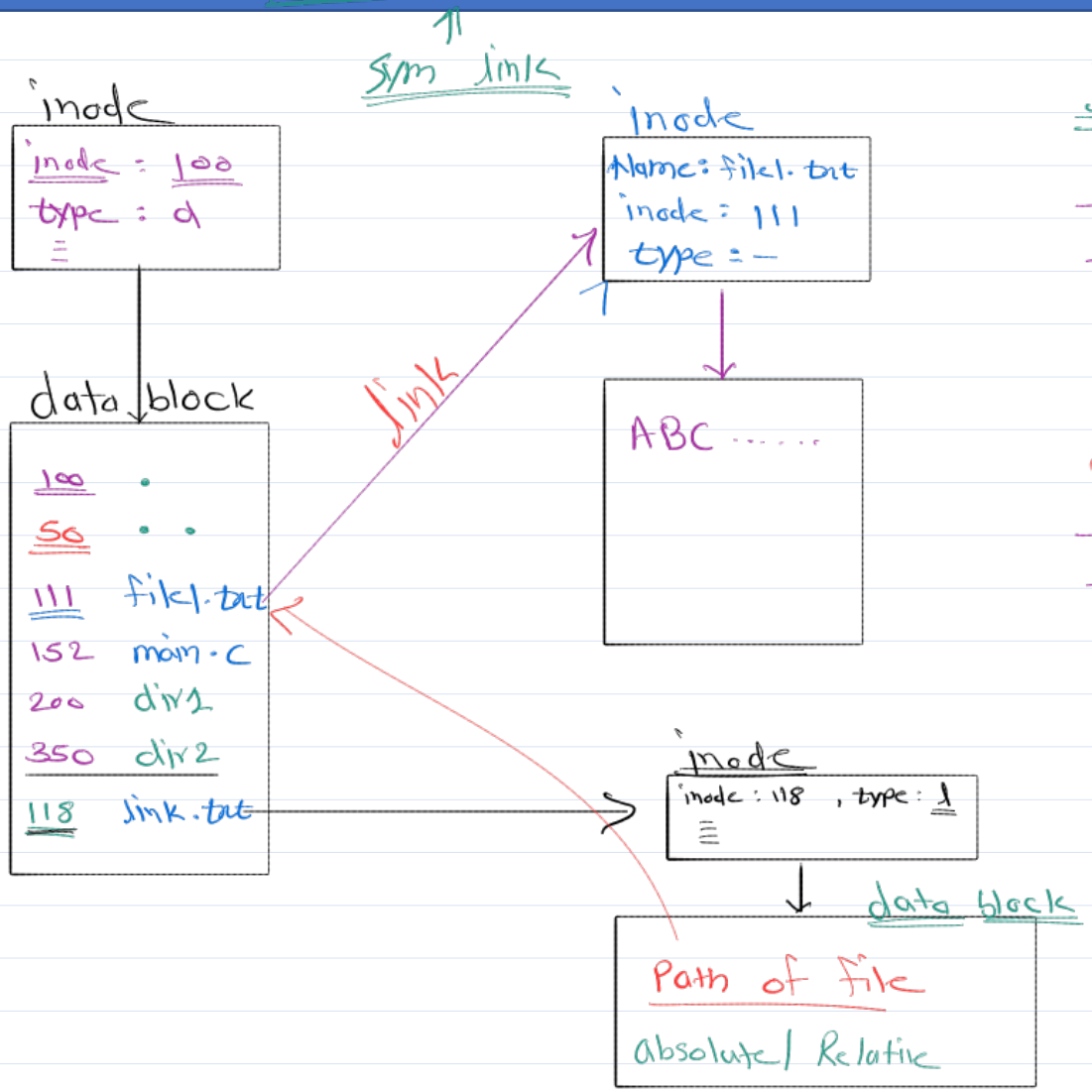
ln file1.txt link.txt

- Create one more entry in directory
- New entry get same inode
- link count get incremented

rm file1.txt

- rm command Remove just dir entry.
- when we delete original file data can access using 2nd link

Symbolic link - ln -s

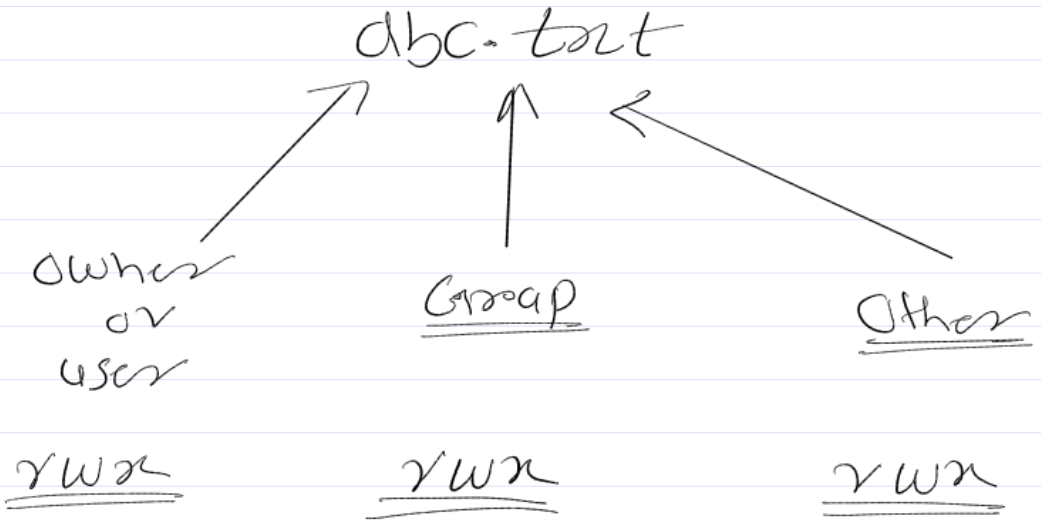


ln -s file1.txt link.txt

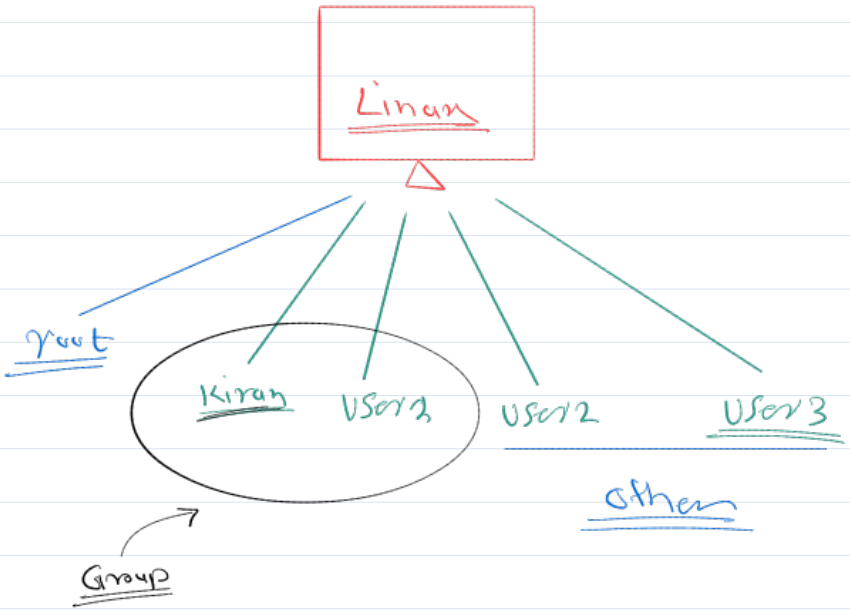
- Create one more new link type file in dir.
- New entry get diff inode
- link count not get incremented

rm file1.txt

- rm command Remove just dir entry.
- when we delete orignal file data can not access using 2nd link, so symbolic link become useless.


abc.txt
owner or user
Group
Other
rwx
rwx
rwx

Chmod :



r	w	x
0	0	0
0	0	1
0	1	0
0	1	1
1	0	0
1	0	1
1	1	0
1	1	1

1 => r/w/x
0 => —

Chmod 000 a.txt
000 000 000
owner Grp other

Chmod 1 4 7 a.txt
001 100 111
owner Grp other
--x r-- rwx

Chmod 5 6 7 a.txt

101 110 111
owner Grp other
r-x rw- rwx

Chmod -rwx filename
Chmod +rwx filename

- => remove , + => add permission

Chmod Command

Permission – mode / access

chmod ± r/w/x filename

+ : add

– : remove

ex.

```
chmod -x sample.txt
chmod +x sample.txt
```

Permission of sample.txt => --x--x--x

chmod u/g/o +r/w/x filename

u – user / owner

g – group

o – others

```
chmod u+rw sample.txt
chmod g+wx sample.txt
chmod o+r sample.txt
```

Syntax:

chmod mode filename

Mode
- Human Format : rw- rw- r--
- Number format : 110 110 100

– Existing mode : rw- rw- r--

– add X to user, remove W from Group

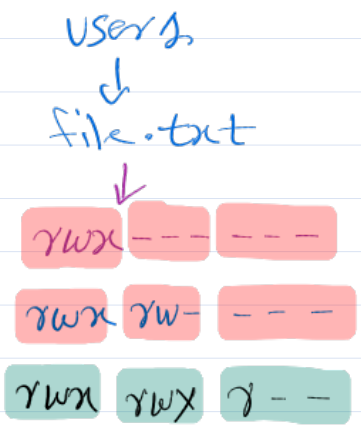
– New mode : rwx r-- r--

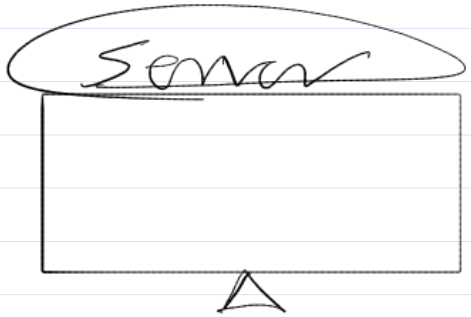
111 100 100

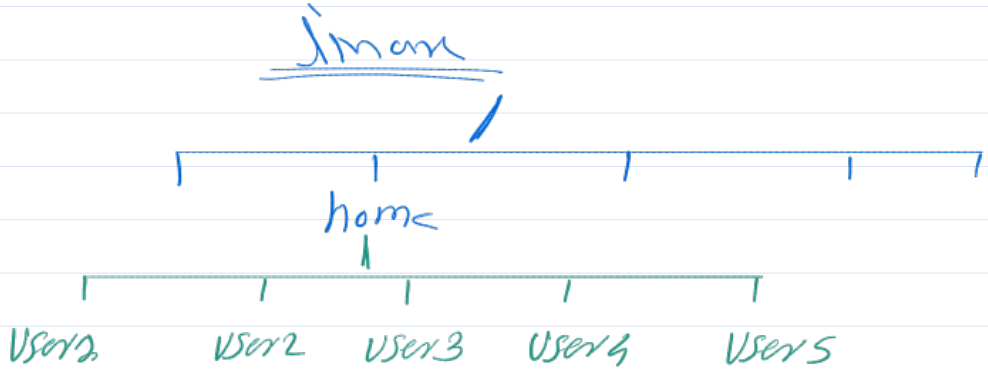
7 4 4

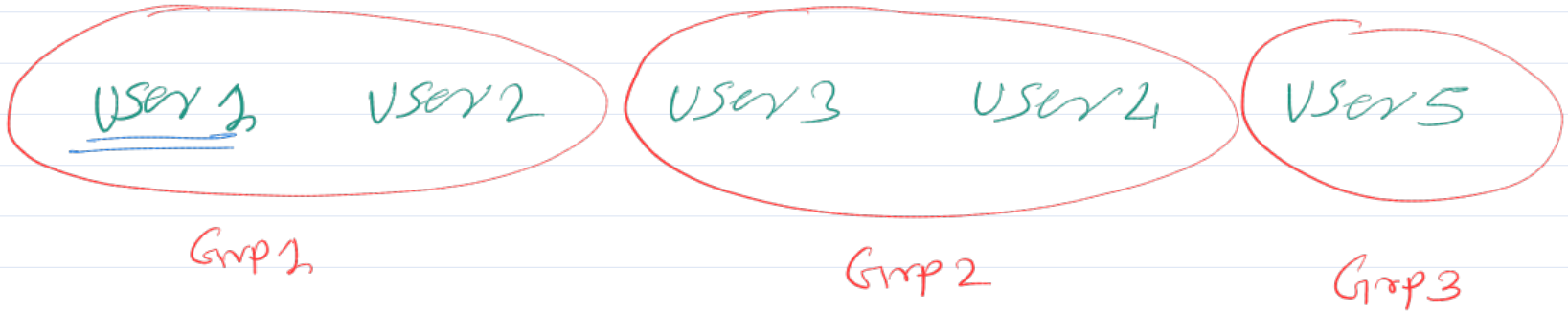
cmd : chmod 744 sample.txt

* Multiuser System





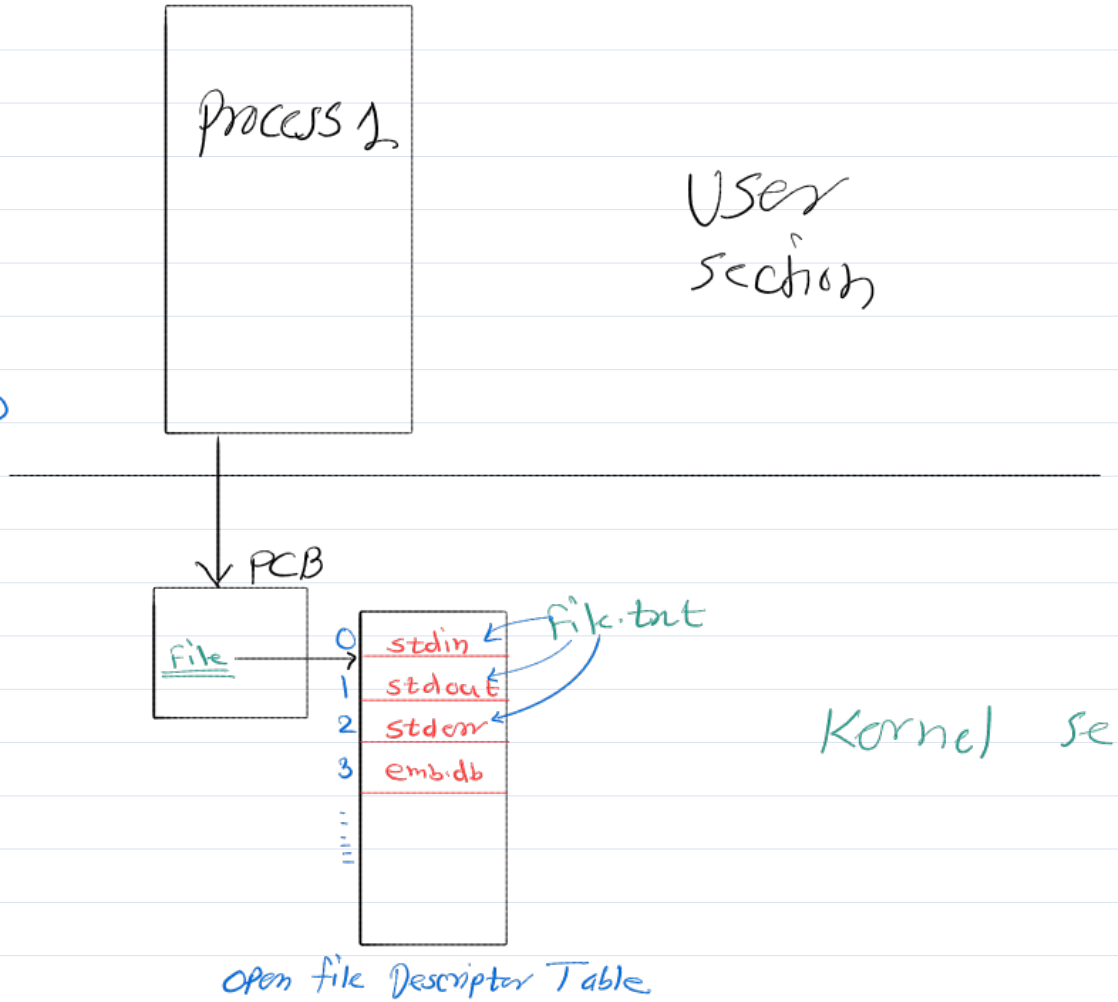




* Using command chmod we can change file permission.

Redirection operator

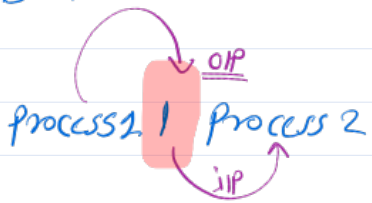
- Each program open three standard file
  1. stdin
  2. stdout
  3. stderr
- Input output is done with these file only.
- But we can change direction of i/p / o/p or error to some user defined file and it is called as "redirection"

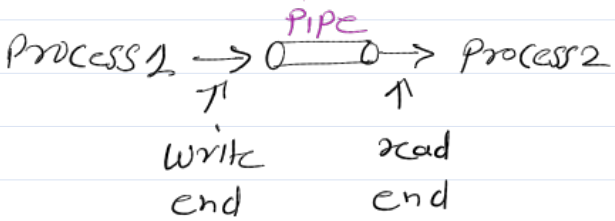


Pipe (1)

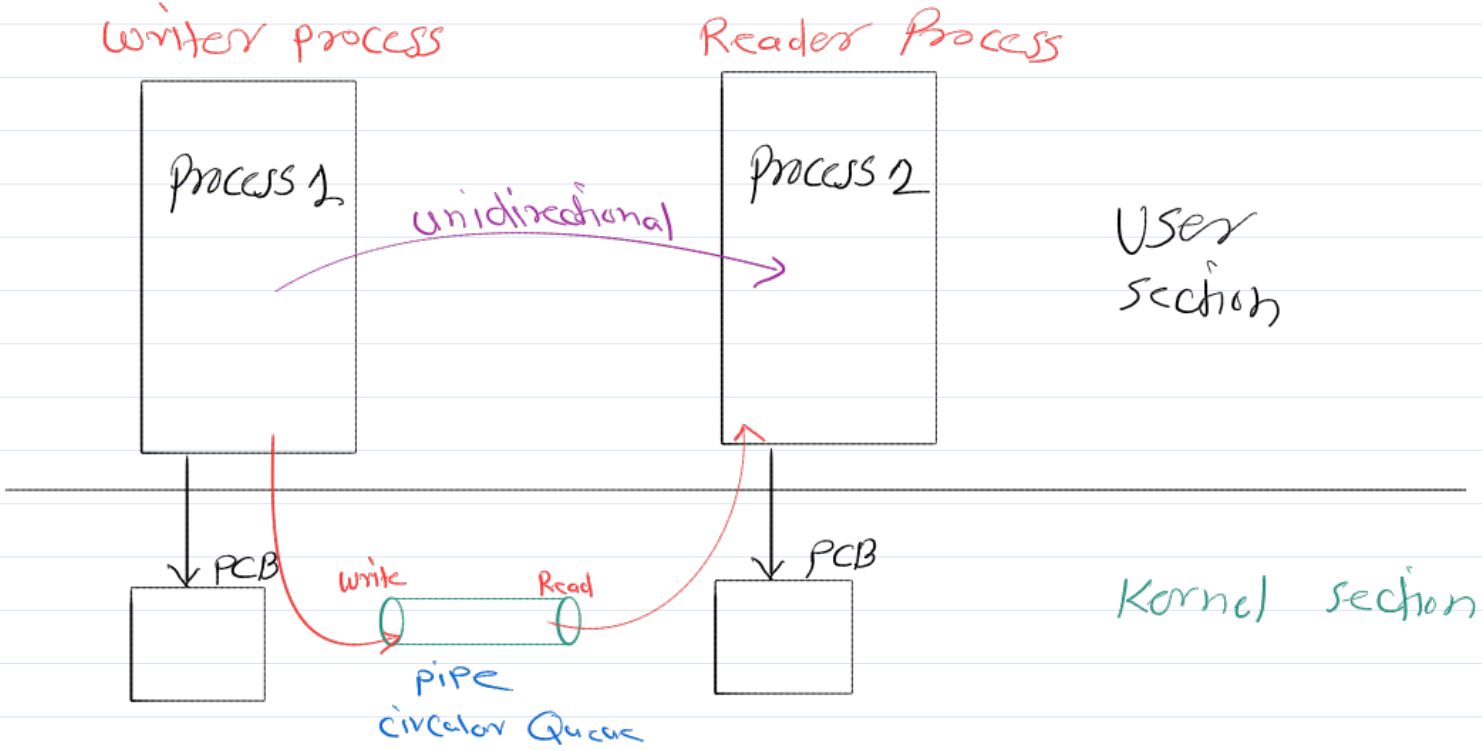
- Pipe used to combine multiple Commands whose output of first Command is given to input of second Command

- Pipe is one of the IPC mechanism.





- Pipes are always unidirectional.



ex. ls -l | wc